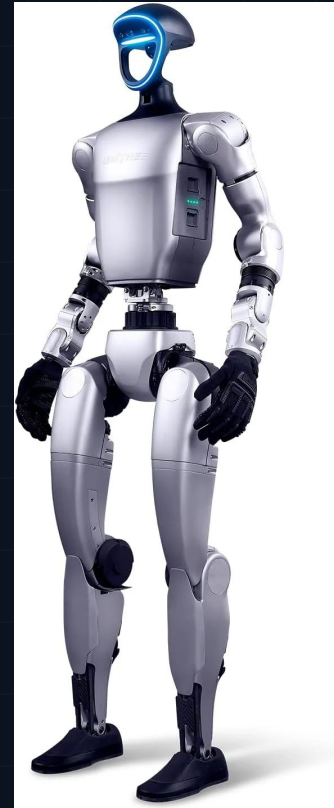


ZERO-SHOT SEGMENTATION FOR ROBOTIC PERCEPTION

Foundation Model Segmentation for Robotic Scenes

Subgroup I2: Zero-Shot Segmentation Benchmark for Robotic Perception (Simulation)
Student: AmirMahdi Matin - 5884715

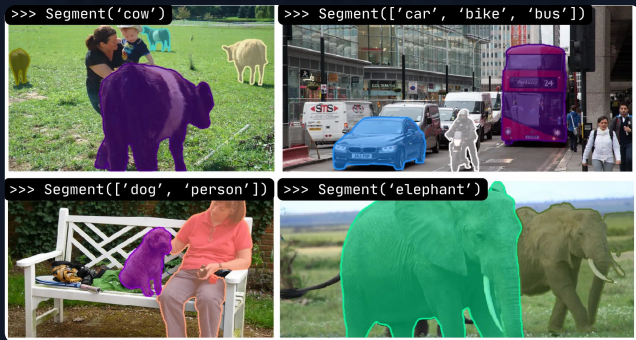
Can promptable foundation models provide reliable zero-shot object masks for robotic perception in challenging simulated scenes?



The Foundation Model Promise vs. the Robotic Reality Gap

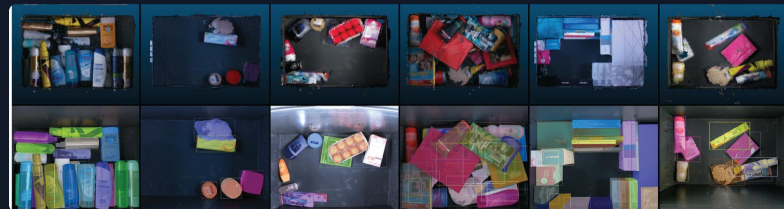
Promptable segmentation works impressively in general images, but robotic scenes add embodiment, clutter, materials, and latency constraints.

Foundation Model Promise



- Point / box / mask prompts
- Broad zero-shot transfer
- High-quality object masks

Robotic Reality Gap



- Clutter and occlusion
- Transparent and reflective materials
- Small parts
- Robot-body visibility
- Real-time constraints

Research need: evaluate segmentation as embodied, action-relevant perception, not only as a generic image task.

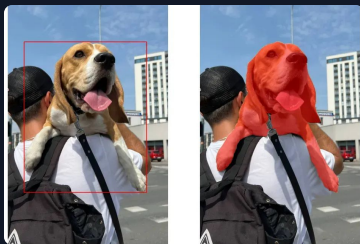
Why Segmentation Matters for Robotic Cognition

Masks become object-level interfaces between perception and action.



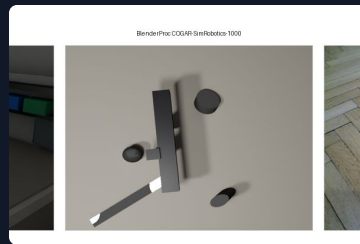
Sensor input

RGB/RGB-D observations
from robot-centered
scenes



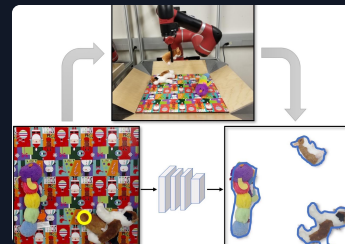
Object mask

Pixel-level foreground
regions for target objects



Object region

Object candidates for
tracking, grasping, or
inspection



Task reasoning

Action selection,
manipulation,
navigation, and
monitoring

Segmentation is a cognitive interface: bad masks can become bad grasps, unsafe navigation, or broken object tracking.

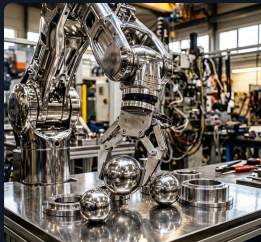
Research Problem: Robotic Scenes Are Visually Hostile

Ordinary segmentation benchmarks understate embodied failure conditions.



Transparent glass

Weak boundaries and background pass-through



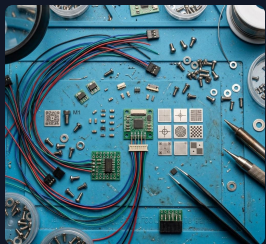
Reflective materials

Specular highlights confuse edge cues



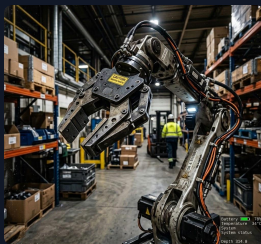
Dense clutter

Object merging and over-segmentation



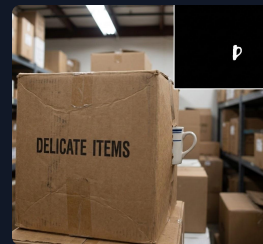
Small parts

Tiny targets are easily missed



Robot visibility

Self-robot pixels create false positives



Occlusion gap

Incomplete masks under occlusion

The benchmark stresses transparent, reflective, occluded, small, dynamic, robot-centered, and cluttered conditions.

State of the Art: Evolution of Segmentation Models

The field has shifted from task-specific supervised models to promptable foundation segmentation and lightweight deployment variants.

Classical / Supervised

Task-specific training on labeled data

- Mask R-CNN
- DeepLabV3+
- YOLOv8-seg



Promptable Foundation

Point, box, mask prompts; SAM2 extends to video

- SAM
- SAM2

Note: SAM3 adds text & exemplar prompting



Lightweight Edge

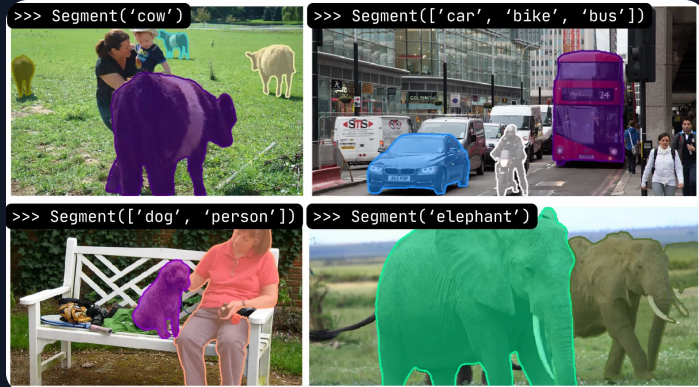
Lower latency and resource requirements

- FastSAM
- MobileSAM
- EfficientSAM

Zero-shot visual generality does not guarantee safe embodied robotic perception.

Research Gap: General Segmentation $\neq$ Robotic Perception

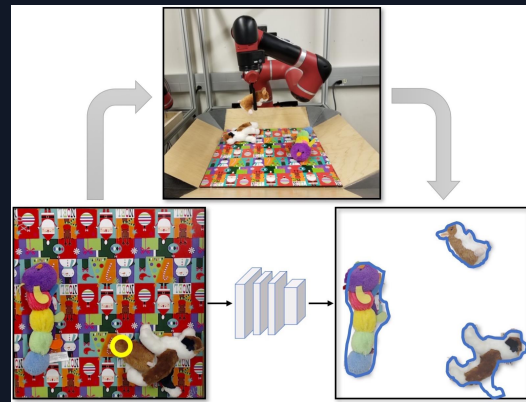
Robotic perception is situated, action-relevant, and time-constrained.



Generic image segmentation

image $\rightarrow$ mask

$\neq$



Robotic cognitive perception

sensor $\rightarrow$ attention $\rightarrow$ mask $\rightarrow$ object state $\rightarrow$ action

Contribution: a controlled benchmark linking mask quality, challenge robustness, speed, and model-selection logic for robotic use cases.

Research Formulation: Four Testable Imperatives

An empirical benchmark of quality, robustness, deployability, and recommendation logic.



1. Datasets

Isaac G1
BlenderProc
OCID



2. Models + prompts

SAM-family
point / box / auto
baselines



3. Metrics

mIoU · Boundary F1
Mask AP
FPS



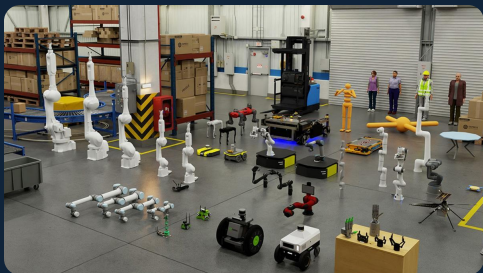
4. Interpretation

failure modes
COGAR mapping
recommendations

Main hypotheses: box prompts outperform point/automatic masks; heavy SAM-family models maximize quality but cost speed; lightweight variants improve deployability with robustness trade-offs.

Dataset and Simulation Design

Simulation enables controlled rare-case evaluation; OCID adds real clutter reference.

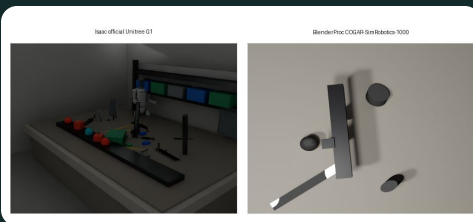


SYNTHETIC **CONTROLLED**

Isaac official Unitree G1

Isaac Sim
1,000 images

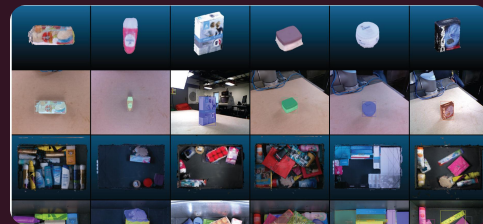
Robot-centered simulation benchmark



BlenderProc COGAR-SimRobotics-1000

Synthetic / BlenderProc
1000 images

Secondary controlled benchmark



OCID

Real RGB-D clutter

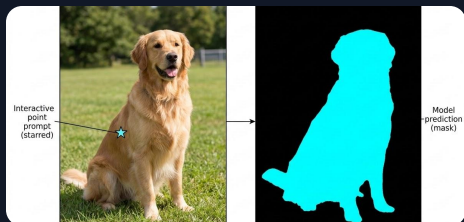
2390 images

Real-world domain-gap reference

Challenge groups: reflective, transparent, occluded, small-part, dynamic, and cluttered robot scenes.

Benchmark Protocol: Prompts, Models, Metrics

Oracle-prompt evaluation isolates mask quality when a target cue is available.



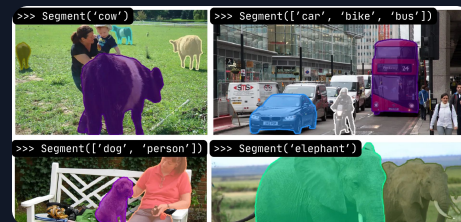
Point prompt

Target cue from one pixel



Box prompt

Target cue from bounding box



Automatic masks

Model proposes masks without a target cue

Models

SAM ViT-H/B · SAM2 · FastSAM · MobileSAM · EfficientSAM · YOLOv8-seg · Mask R-CNN · DeepLabV3+

Metrics

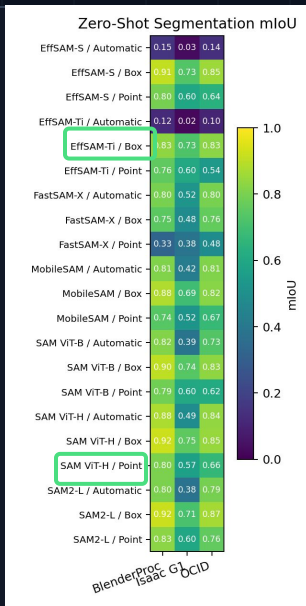
mIoU · Boundary F1 · Mask AP/AP50/AP75 · per-category IoU · GPU/CPU FPS

Boundary Note

Point/box prompts are oracle prompts; real robots need detector, tracker, planner, human instruction, or task prior.

Mask Quality: Strong, but Dataset-Dependent

Prompted foundation models can produce strong masks, but performance varies across robotic datasets.



0.923

SAM ViT-H box
BlenderProc

0.752

SAM ViT-H box
Isaac G1

0.883

MobileSAM box
BlenderProc

0.824

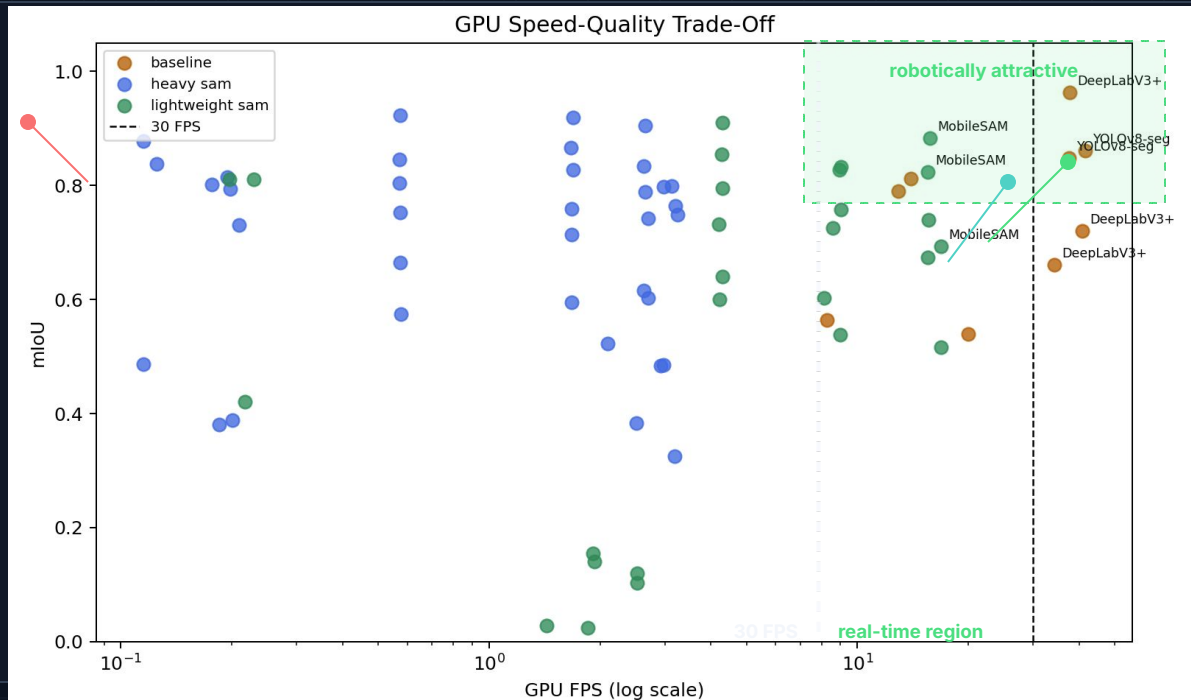
MobileSAM box
OCID

Box prompts are strong, but dataset difficulty changes the ranking.

High mIoU is promising, but robotic value also depends on robustness and speed.

The Latency Reality: Accuracy Is Not Enough

Robotic deployment requires joint evaluation of mask quality and inference speed.



0.574 FPS

SAM ViT-H box

High fidelity, poor latency

41.647 FPS

YOLOv8-seg

Real-time capable

37.811 FPS

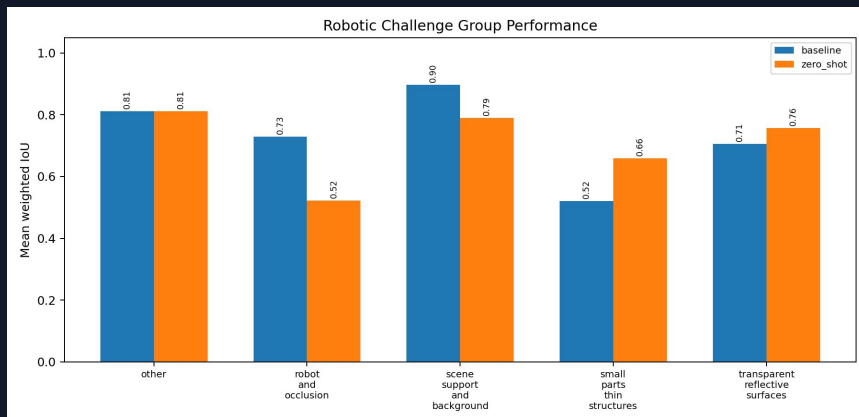
DeepLabV3+ on OCID

Fast supervised baseline

A high-quality mask is not robotically useful if it arrives too late for the perception-action loop.

Challenge-Group Vulnerabilities

Average scores hide material-specific and embodiment-specific failures.



Transparent Objects

Weak/missing boundaries → unreliable grasp estimation

Reflective Surfaces

Highlight-edge confusion → false object edges

Small Parts

Missed regions → target disappears from system

Occlusion

Incomplete masks → unstable tracking

Dynamic Body

Self-scene confusion → robot body confused with target

The failure category can matter more than the average leaderboard rank. Per-condition evaluation reveals dangerous robotic failures hidden in average scores.

Failure Mode Analysis: Where Models Break

Qualitative failures explain the robotic risk behind the metrics.

MISSED TARGET

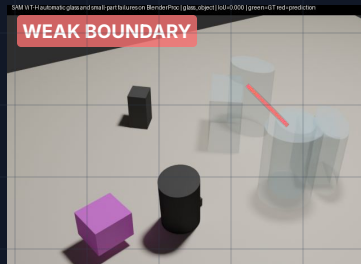


Small part failure

Failure: tiny screw / connector missed

Risk: target may disappear from object representation

WEAK BOUNDARY

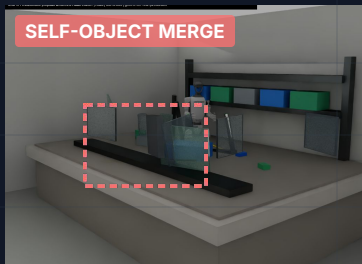


Transparent object failure

Failure: weak boundary; background bleed

Risk: unreliable grasping or collision estimation

SELF-OBJECT MERGE



Robot-body interference

Failure: self-robot pixels merge with scene

Risk: robot confuses own body with targets

OBJECT MERGE



Real clutter and occlusion

Failure: nearby objects merge; incomplete masks

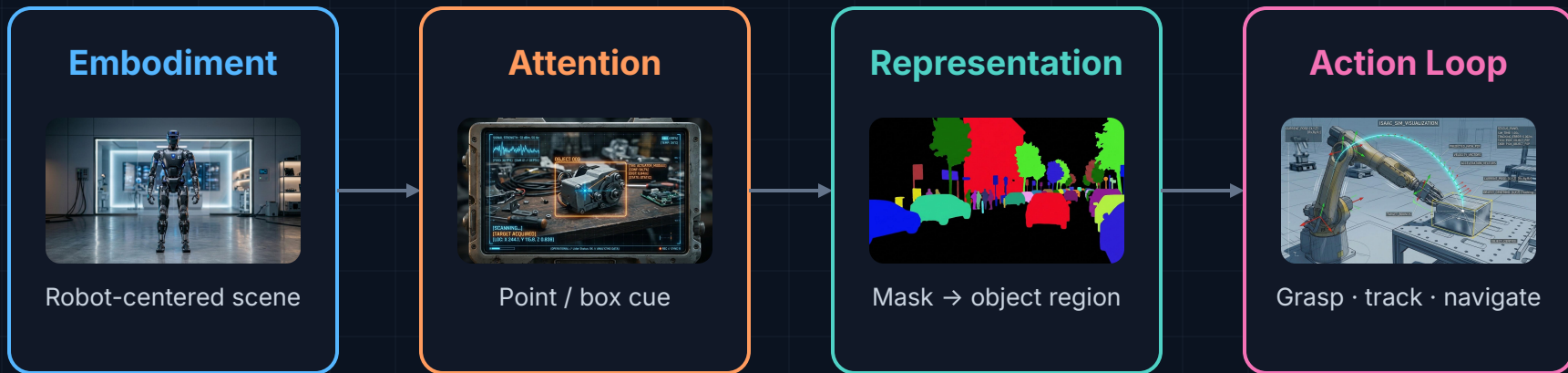
Risk: tracking, grasping, or counting failure

Small mask errors can become large robotic errors: missed grasps, unsafe motion, broken tracking, or incorrect object memory.

Cognitive Approach: Segmentation as a Perception Module

Segmentation links visual input, attention, object representation, and action selection.

Perception-to-action flow



COGAR link: Masks act as perceptual structures for attention, memory, object reasoning, and action selection.

Conclusions & Model-Selection Blueprint

The best segmentation model depends on the robotic task, prompt availability, robustness requirement, and latency budget.

Robotic Scenario Recommended Direction Why It Works: Required Caution

Offline / High Mask Quality **SAM / SAM2 (box prompts)** Strongest mask fidelity Slow, needs reliable cue

Prompt-Guided Manipulation **Box-prompted SAM models** Strong target masks Upstream detector required

Real-Time GPU Perception **YOLOv8-seg / DeepLabV3+ High FPS** (with data) Weak zero-shot transfer

Edge Deployment **MobileSAM / EfficientSAM** Quality-speed balance Drops in clutter / small parts

Prompts are prerequisites

High performance often assumes an upstream cue.

Physics defeats averages

Transparent, reflective, and small objects remain difficult.

Speed is a constraint

Latency limits closed-loop perception and action.

Foundation segmentation is useful for robotic perception, but only when prompt source, failure checks, and latency constraints are designed into the perception-action loop.

Thank You for your attention.

COGNITIVE ARCHITECTURES FOR ROBOTICS

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